VITALITY 15

Papaverine

Alkaloid obtained from opium or made synthetically; relaxes smooth muscle directly



CLASS
Alkaloid (smooth-muscle relaxant)

STUDIED FOR
Erectile dysfunction (injected)

HUMAN EVIDENCE
Small human studies

US STATUS
Not FDA approved; sold as unapproved Rx

STUDIED FOR

♂ **Erections**
HUMAN

Label: not for penile injection. The injection label says it is not indicated for impotence, and that penile injection has caused persistent priapism.

HOW IT WORKS IN THE BODY

Papaverine ▶ Smooth muscle ▶ Arteries widen ▶ Penile blood flow

- **Relaxes smooth muscle directly,** especially in large arteries (label)
- Acts on **blood vessels** and the gut, bile and urinary tracts
- **Minimal effects on the brain,** per the label

EVIDENCE & WHAT'S KNOWN

EVIDENCE

SMALL HUMAN
LEVEL 3 OF 5

- 1982 letter first reported penile injection
- Small comparison studies: 60 and 180 men
- No large placebo-controlled ED trial

NOT YET ESTABLISHED

- Half-life (estimates vary)
- A standard Trimix recipe

QUICK PRACTICAL INFO

HALF-LIFE
Not established; label says estimates vary widely

ROUTES STUDIED
Injection (label); penile injection is off-label

PHARMACOKINETICS
Liver metabolism; about 90% bound to blood proteins

SOLD AS
Unapproved Rx injection; compounded Bimix and Trimix

MONITORED IN TRIALS
Liver tests, eosinophils, prolonged erections

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Not approved; marketed as an unapproved drug
FDA	Label: not for injection into the penis
WADA	Not on the 2026 list

REFERENCES

1. American Regent, papaverine injection label (DailyMed), Jul 2026
2. Bechara et al., J Urol, papaverine-phentolamine vs alprostadil, Jun 1997
3. Seyam et al., Int J Impot Res, Trimix vs alprostadil, Jul 2005

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